

Summer Internship Report



Data Science and Analytics Group (DSAG)

Project Title:

Sourcing Quality Interactive Dashboard Preparation – Retail

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To my peers, family, and friends, I express my heartfelt thanks for their encouragement and moral support. This internship has been a rewarding experience, and I am thankful for the collective efforts that made it possible.

Rick Sundar Jena
Summer Intern

ABSTRACT

The present report outlines the work undertaken by the undersigned during a **2-M Internship** at **ICICI Bank Ltd.**, conducted from **1st June 2025 to 31st July 2025**, under the **DSAG (Data Science and Analytics Group)**. The internship was carried out in **offline mode** at the ICICI Bank Office, **Plot Number 06, 6th Floor, Paras Downtown Centre, Sector 53, Gurugram, Haryana – 122002**.

The primary project assigned during this internship was titled “**Sourcing Quality Interactive Dashboard Preparation – Retail**”. The focus was on building a robust and insightful **Power BI Dashboard** to help ICICI Bank evaluate the profitability and sourcing quality of its **auto loan portfolio**.

The main objectives of the project were:

- Profitability Analysis
- Sourcing & Default Analysis
- Future Profit Forecasting
- What-If Analysis

This internship not only enhanced my technical skills in **data visualization, data modeling**, and **business analytics**, but also deepened my understanding of **real-world banking operations, risk evaluation**, and **profitability analysis** in the financial sector.

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1. INTRODUCTION

This report provides a comprehensive documentation of the internship project titled “***Sourcing Quality Interactive Dashboard – Retail***”, which was undertaken as part of a two-month summer internship program at ICICI Bank Ltd. from 1st June 2025 to 31st July 2025. The project was carried out under the Data Science and Analytics Group (DSAG), which plays a critical role in enabling data-driven decision-making within the bank’s operations.

The core focus of the project was on the **Auto loans’ portfolio** within **ICICI Bank’s retail lending segment**. Auto loan products form a significant part of the bank’s asset base and are critical for long-term profitability and risk management. The internship aimed to explore profitability, sourcing quality, and forecasting strategies associated with this portfolio using simulated yet realistic datasets, and to convert analytical models into a practical business dashboard.

The primary goal was to build an end-to-end dashboard using Microsoft Power BI that could serve as a decision-making tool for retail product managers and strategy teams. The project has been completed with the help of simulated datasets of over 100K Customers under retail loan segment.

Retail lending is a core vertical within ICICI Bank, involving the disbursement of home loans, auto loans, personal loans, and other credit instruments to individual customers. However, each customer segment brings a different level of profitability and risk. Sourcing quality and customer selection significantly influence a bank’s financial performance. Recognizing the criticality of this domain, the internship project focused on developing a robust framework to track, visualize, and simulate profitability and sourcing dynamics. The main objective was to fulfill all the

requirements of the entitled project and to develop a fully interactive Power BI dashboard which is highly user friendly.

The interactive dashboard developed as part of this project is designed to:

- Analyze loan profitability across various customer segments, including CIBIL score bands, customer profiles, sourcing channels, new/upgrade customers and vehicle segments
- Evaluate sourcing quality, by highlighting default trends using Bad rate at 12 & 24 MOB, Default and NPA rate, across various sourcing financial years
- Forecast future profitability, using historical performance data of the previous eight quarters
- Simulate alternate sourcing scenarios, using What-If parameters to model the impact of share shifts among Channels, CIBIL bands, Profiles, Vehicle Segments etc.

The dashboard is fully dynamic, modular in structure, and built on simulated yet realistic datasets that mimics ICICI Bank's Auto loans portfolio structure. Through the use of DAX (Data Analysis Expressions) and Power BI's visualization tools, the project successfully translated complex profitability calculations into clear, actionable insights.

Each module within the dashboard is built to answer a specific set of business questions. Users can filter data, view trends, and interact with What-If sliders to simulate the impact of changing business strategies.

This report further elaborates on every module, visual component, DAX logic, and KPI used in the dashboard. The goal is to provide a clear, user-friendly manual that enables any stakeholder; whether technical or non-technical, to understand and use the dashboard effectively for business decision-making. This report aids the understandability of the designed dashboard covering each and every aspect of the ***Sourcing Quality Interactive Dashboard – Retail*** in detail.

2. PROJECT OVERVIEW

The Power BI dashboard developed during the internship is designed to serve as a strategic analytical tool for evaluating the auto loans' portfolio performance at ICICI Bank. The goal was not limited to just creating visuals, but to derive meaningful business insights that can support leadership decisions on improving profitability, managing credit risk, and optimizing sourcing strategies.

The goal of the project was to use **data visualization and analytics** to answer critical business questions such as:

- *Which sourcing channels and customer types are profitable?*
- *Are we improving in sourcing better quality customers over time?*
- *What happens if we change the sourcing strategy?*
- *How does profitability look in the future based on current trends?*

The dashboard addresses the following core business objectives in detail:

2.1. Profitability Analysis:

A major objective of the dashboard was to identify customer segments that are contributing positively or negatively to the bank's profitability. In the context of retail lending — and especially in auto loans — profit drivers and cost levers vary widely across different types of customers. Some segments yield consistently high profits due to favorable credit scores, while others pose a high credit risk and generate losses.

To uncover such trends, the dashboard includes a dedicated **Profitability Analysis module** that allows users to analyze Profit Before Tax (PBT); Both cumulative and L4Q in absolute as well as in percentage form across various dimensions such as:

- CIBIL Score Bands (*-1 (No History), Less than 700, 700 to 725, 725 to 750, 750 to 780, 780 to 800, More than 800*)
- Profile (*SAL, SE+OTHERS*)
- Channel (*BRANCH, DSA, OAP*)
- Customer type (*NEW, UPGRADE*)
- Vehicle Segment (*Luxury, Non-Luxury*)

Through slicers, KPI tiles, and dynamic visuals, business users can instantly view which combination of segments are delivering high returns, and which are more profitable. This insight is crucial for defining more effective sourcing criteria and prioritizing high-yield customer segments going forward.

Additionally, revenue and cost components such as **interest income, fees, recovery, BOC, ALMG, cost of funds (COF), credit cost(CC), DMA, DRE, OPEX** are analyzed to offer a breakdown of how profits are being generated. These measures are calculated using custom **DAX logic** to ensure real-time updates and dynamic filtering using various slicers that has been added in the slicer panel.

2.2. Sourcing Quality and Default Analysis:

Another central objective of the dashboard is to evaluate **sourcing quality**, which refers to the long-term sustainability and creditworthiness of customers being on boarded through different sourcing channels.

High disbursal volume does not always equate to quality sourcing — if the customers sourced are prone to early-stage delinquencies or default risk, then even a large portfolio can lead to future

losses. Therefore, it is essential for the bank to monitor the **Bad rates, Default & NPA rate**, and segment-specific risk performance across sourcing categories.

To address this, the dashboard includes a **Sourcing Quality module** that shows:

- **Bad Rate @ 12 & 24 MOB (Months on Book)**
- **NPA Rate Trend**
- **Default Rate**

This module is especially useful for the credit and risk analytics teams, as it provides a dynamic view of credit performance vs sourcing strategy, improves underwriting and helps identify red flags early. Business teams can also assess which channel of sourcing is bringing in more default-prone customers, and adjust their partner allocations accordingly.

The calculations are driven through well-defined DAX measures and are visualized using dynamic charts, trend lines which are responsive to multiple slicers incorporated in the slicer panel. The detailed explanation for each measure used has been discussed later in the report. The DAX code used for calculation has been added further for reference purpose. In addition to this, the dashboard snapshots have been added for better visualization.

2.3 Future Profitability Forecasting:

The third core objective was to develop a **forward-looking forecasting tool** that can project **future profitability trends** using past performance data.

Forecasting is essential for strategic planning, as it enables the business to assess whether their current sourcing approach and customer mix will continue to yield positive returns over the next few quarters.

In this module, historical PBT trends — typically from the **last 8 quarters** — are used to forecast **PBT values for the next quarters**. The forecast logic is based on **Built-in time series forecasting tools** within Power BI which uses the *ETS (Exponential Triple Smoothing)* model, that analyses the overall pattern over the last 8 quarter data and sees the level, trend and seasonality, thus predicting the upcoming values of profit % for the next quarters.

The forecasted values are displayed using time-series line charts, with shaded areas representing the future projections. These visuals help business teams visualize **seasonality patterns**, **Quarter-On-Quarter growth**, and anticipate any profitability dips ahead of time.

This module is interactive and responds to filters such as loan customer segments, sourcing channel and other parameters, thus making it useful for both macro-level planning and review.

2.4. What-IF Analysis:

One of the **most powerful features** of the dashboard is the **What-If Simulation module**, which allows business users to simulate sourcing strategy changes, observe the **real-time impact** on profitability metrics like PBT %, Internal Rate of Return (IRR%), fee income, interest income and recovery as well as cost components (CC, OPEX, DMA, DRE) and analyze the change in each single component under the impact table created.

The sourcing strategies can be adjusted by modifying the proportion of customers coming from different segments. Users can test the impact of increasing or decreasing the share of a specific channel, CIBIL band, Profile Band, New/Upgrade customer, Vehicle segment and Pre-Approved (PA)/ Non-PA loan customers, and immediately see how that affects the dashboard's profitability calculations.

There are six simulation categories built into this module:

1. **Channel-Based Simulation:** Adjust the share of **B R A N C H**, **DSA** and **OAP** sourcing parameters and see how profit metrics change.
2. **Customer Type Simulation:** Modify the share between **New** and **Upgrade** customers and evaluate segment-level impact.
3. **CIBIL Mix Simulation:** Model the impact of sourcing more high-**CIBIL** customers and fewer low-score customers, or vice versa.
4. **Profile Band Simulation:** Adjust the sourcing share between **Salaried (SAL)** and **Self-Employed+Others (SE+OTHERS)** to simulate how customer affordability influences PBT.
5. **Vehicle Segment Simulation:** Modify the share between **Luxury** and **Non-Luxury** car buyers and evaluate impact
6. **PA/NON PA Simulation:** Modifying the share of **Pre-approved** loans and see how it impacts the profitability

Each simulation uses **What-If Parameters** and custom DAX logic to calculate the **Simulated PBT vs Base PBT**, along with visual cards showing % change and financial gain/loss. This makes the dashboard not just analytical but also **prescriptive** — helping decision-makers answer “**What will happen if we change our approach?**”

The What-If analysis is divided into **two** major parts; a **Unified Simulation** and an **Individual Simulation**.

The **Unified Simulation**, as the name suggests is a What-If simulation with all the 6 categories of simulation impacting the **PBT, IRR, Recovery, Revenue (Fee and Interest Income)** and cost (**CC, DRE, DMA, OPEX**) components together on a single page. However, the Individual Simulation, shows impact of all the six categories separately in separate pages for more in-depth

insights.

This module equips **dynamic simulation engine** that can guide future planning, sourcing optimization, and scenario-based business strategy testing.

The module also includes a dynamic card that validates the user's inputs. If the input is equal to 100%, it returns a VALID confirmation. Whereas, whenever the inputs exceed 100%, it returns an INVALID confirmation.

3. METHODOLOGY

This section outlines the complete methodology followed for developing the Power BI dashboard and the tools used at each stage of the project lifecycle. The methodology was designed to simulate real-world business scenarios within **ICICI Bank's Auto loans portfolio**, ensuring that the dashboard could deliver practical insights while remaining scalable, modular, and intuitive.

3.1. Methodology Followed:

To achieve the project's objectives, the project was divided into multiple logical phases:

- Understanding **key banking variables** like CIBIL, SOL-ID, LAN -ID, UCIC, PA Flag, PBT CUMM, PBT L4Q, DPD(Current), 90+ at 12 and 24 MOB, 30+ at 12 and 24 MOB and multiple revenue and cost components. Based on this financial understanding, creating a realistic variable list under the guidance of mentor.
- Generating a simulated dataset using Excel and python to simulate ICICI Bank's auto loans' vehicle segments and DPD buckets. The dataset was refined for logical consistency.
- Coding **DAX (Data Analysis Expressions)** formulas to calculate: Profit Before Tax (PBT L4Q%, PBT CUMM %) and its components, Bad Rate @ 12 MOB, NPA Rate, Default Rate, multiple What-If Simulation Metrics and other related crucial parameters were calculated logically via DAX Measure in Power BI.
- Designing a modular, objective-wise dashboard layout using **Microsoft Power BI**. Each core module was represented using a separate page.
- Creating What-If Parameters to allow dynamic simulation of sourcing share between:

1. Sourcing channels
2. Sourcing CIBILs

3. Profile Bands

4. New/Upgrade type of customers

5. Vehicle Segments

6. Pre-Approved (PA)/ Non-PA

3.2 Tools and Technologies Used:

- Microsoft Power BI Core dashboard development, interactive visuals, slicers, DAX
- Microsoft Excel Data simulation, variable design, preliminary calculations
- Python (basic) Used for dataset generation and logic validation
- DAX (Power BI Language) Used extensively to compute dynamic KPIs, simulations, and conditions

These tools were selected to strike a balance between simplicity (Excel), data visualization (Power BI), and logic control (DAX), ensuring the dashboard was technically strong and business-ready.

4. DASHBOARD STRUCTURE

The dashboard created during this internship is structured in a modular and objective-oriented manner. Each page is carefully designed to address a specific business question and allows users to explore insights interactively, compare results, and simulate future scenarios. This structure improves usability for both technical and non-technical stakeholders, ensuring clarity, navigability, and impact.

The dashboard is well organized for analysis of each objective in-depth and follows easy navigation and user friendly experience. Each page corresponds to one of the main project objectives:

- i. Homepage (Navigation Page)
- ii. Profitability Analysis
- iii. Sourcing Quality & Default Trend,
- iv. Future Forecasting
- v. What-If Simulation (Unified) *and further bifurcated into individual What-If analyses*

Let us now have a look at each segment of the designed Power BI Dashboard one by one:

Page 1: Homepage (Navigation Hub)

This is the welcome screen and central navigation page of the dashboard. It is designed for quick and intuitive access to all other modules.

- **Features:**
 - **Navigation buttons:** Each button, redirects the user to a different objective (*Profitability, Sourcing Quality, Forecasting, What-If*)
 - **Top KPI Cards:** Present a summary of key metrics of the portfolio such as Total Loans

Disbursed, Average CIBIL Score, Average IRR, Profit (Cumulative), PBT% (Current and L4Q), Revenue (Cumulative, %Current), Costs (Cumulative, %Current), Disbursal Amount (Cumulative) and TPOS (Cumulative)

- **Slicers** with Disbursal Year/Month/Quarters, Live/Close, PA/Non-PA, Profile bands, New/Upgrade, Vehicle Segments, Sourcing Channels, DPD Buckets, CIBIL bands, Vehicle Manufacturers and States for more comprehensive overview of the portfolio
- **Purpose:**
 - Acts as clean entry point for users
 - Provides a purpose for viewing the dashboard
 - Helps users jump directly to their area of interest

Page 2: Profitability Analysis

This module focuses on identifying which customer segments are more profitable in the Last 4 Quarters (L4Q) and helps tackle questions like:

- Which customer profile (***SAL vs SE+OTHERS***) or (***NEW vs UPGRADE***) or ***CIBIL bands*** give superior returns?
- Which channels (***BRANCH/DSA/OAP***) or (***PA/Non-PA***) bring for loss-making customers?
- Which states contribute more to the profitability of the bank?
- Which vehicle manufacturers or segments are more profitable to source?
- Which type of customers have given the highest profit/loss to the bank?
- What is the total profit earned till date by the bank?

- **Features:**
 - **Dynamic Table** with All the Revenue, Recovery and Costs components (as a % of TPOS), PBT%, Profit per loan, Total TPOS with slicer for Profile Bands, Channels, New/Upgrade/States/ CIBIL Bands etc. for more comprehensive insights
 - **Column Chart** with PBT (%), Revenue (%) and Costs (%) mapped against Disbursal Date
 - **Slicers** for Disbursal Year/Month/Quarters, Live/Close, PA/Non-PA, Profile bands, New/Upgrade, Vehicle Segments, Sourcing Channels, DPD Buckets, CIBIL bands, Vehicle

Manufacturers and States

- **Summary Cards** of key KPIs
- **Purpose:**
 - Helps the user identify where to channel sourcing efforts to maximize returns.

Page 3: Sourcing Quality and Default Analysis

This module is used to evaluate credit performance and sourcing quality – whether the sourced customers are actually paying back or defaulting early.

- **Key Metrics Analyzed:**
 - **Bad Rates @ 12 & 24 MOB:** Calculated using *30+ & 90+ Amount at 12 & 24 MOB* and *Disbursal Amount*
 - **NPA Rate** as of Q4 FY25: *TPOS of customers with DPD>90+days* and *Total Disbursal Amount*
 - **Default Rate:** *TPOS of customers with DPD >0days* and *Total Disbursal Amount*
- **Key Visuals:**
 - **Dynamic Line Chart** showing *Bad Rates 30+ & 90+ at 12 & 24 MOB*, *Default Rate* and *NPA Rate* trend across *Disbursal date* for vintage analysis
 - **Dynamic Table** with all the *Bad Rates*, *Default Rate* and *NPA Rate* across PA/Non-PA, Profile bands, New/Upgrade, Vehicle Segments, Sourcing Channels, DPD Buckets, CIBIL bands, Vehicle Manufacturers and States
 - **Summary Cards** for Key KPIs
 - **Slicers** for Disbursal Year/Month/Quarters, Live/Close, PA/Non-PA, Profile bands, New/Upgrade, Vehicle Segments, Sourcing Channels, DPD Buckets, CIBIL bands, Vehicle Manufacturers and States

- **Purpose:**

- Anticipating the risk across various sourcing parameters
- Comparing loans on the basis of credit quality

Page 4: Future Profit Forecasting

This module addresses the strategic need for analyzing the current profitability trend for the past 8 Quarters and forecasting the next quarters on the existing trend. It analyzes where the business is headed in terms of financial returns.

- **Forecasting Logic:**

- Uses historical PBT data for the trailing 8 Quarters (Q1 FY24 to Q4 FY25)
- Predicts next 2 Quarters PBT using ETS (***forecasting model that analyzes Level, Trend and Seasonality***) and on the basis of past trend, it forecasts the values

- **Key Components:**

- **Line Graph** of historical trend followed by forecast trend
- **Slicers** all the sourcing parameters

- **Purpose:**

- Anticipate Quarter on Quarter (QoQ) profit % and trend analysis
- Plan sourcing or risk targets based on forecasted values

Page 5: What-IF Scenario (Unified Simulation)

This is the most strategic and interactive part of the dashboard. It enables the user to simulate changes

in mix of sourcing parameters and simultaneously analyze the impact on key profitability metrics.

- **The Unified Simulation includes the following components:**
 - **Sliders:** for CIBIL Bands, Channels, Profile Bands, New/Upgrade customers, Vehicle Segments and PA sourcing for mixing of share values to simulate What-if changes
 - **Share Tables:** with base share and simulated share of all the sourcing parameters to see changes in the mix values manually
 - **Base vs Simulated Value Tables:** with base values, simulated values and %changes for PBT%, revenue, recovery and cost components to analyze the impact of mix change
 - **Reset Button** for setting values automatically to base values
 - **Navigation button for individual simulations:** for more in-depth analysis per sourcing parameter
 - **Dynamic Input Validation Card:** to ensure the user knows if the inputs are equal to 100% or exceeds it.

Page 6-11: What-IF Scenario (Individual Simulation)

This module delves into more in-depth analysis of what-if simulation for each sourcing parameter, for more informed changes in key KPIs

- **Page 6: Channel Mix:**
 - **Sliders** for adjusting mix values for **channel** (*BRANCH, DSA, OAP*)
 - **Simulated Values Table** for viewing the changes in simulated values across the channels for PBT, recovery, revenue and cost components
 - **Base vs Simulated Values Table** for analyzing the changes in the simulation values as compared to the base values across the PBT, recovery, revenue and cost components
 - **Share Table** for analyzing the simulated shares vs the base shares
 - **Summary Cards** for Key KPIs
 - **Dynamic Text Card** for showing change in PBT% and the highest share channel
- **Page 7: CIBIL Mix:**

- **Sliders** for adjusting mix values of **CIBIL Bands** (-1 (*No History*), *less than 700*, 700-725, 725-750, 750-780, 780-800, *more than 800*)
- **Simulated Values Table** for viewing the changes in simulated values across the CIBIL bands for PBT, recovery, revenue and cost components
- **Base vs Simulated Values Table** for analyzing the changes in the simulation values as compared to the base values across the PBT, recovery, revenue and cost components
- **Share Table** for analyzing the simulated shares vs the base shares
- **Summary Cards** for Key KPIs
- **Dynamic Text Card** for showing change in PBT% and the highest share CIBIL band

- **Page 8: Profile Mix:**

- **Sliders** for adjusting mix values for **Profile bands** (*SAL, SE+OTHERS*)
- **Simulated Values Table** for viewing the changes in simulated values across the profiles for PBT, recovery, revenue and cost components
- **Base vs Simulated Values Table** for analyzing the changes in the simulation values as compared to the base values across the PBT, recovery, revenue and cost components
- **Share Table** for analyzing the simulated shares vs the base shares
- **Summary Cards** for Key KPIs
- **Dynamic Text Card** for showing change in PBT% and the highest share profile band

- **Page 9: NEW/UPGRADE Mix:**

- **Sliders** for adjusting mix values for **New/Upgrade** customers
- **Simulated Values Table** for viewing the changes in simulated values across the type of customers for PBT, recovery, revenue and cost components
- **Base vs Simulated Values Table** for analyzing the changes in the simulation values as compared to the base values across the PBT, recovery, revenue and cost components
- **Share Table** for analyzing the simulated shares vs the base shares
- **Summary Cards** for Key KPIs
- **Dynamic Text Card** for showing change in PBT% and the highest share customer type

- **Page 10: Segment Mix:**

- **Sliders** for adjusting mix values for **vehicle segments** (*Luxury, Non-Luxury*)
- **Simulated Values Table** for viewing the changes in simulated values across the segments for PBT, recovery, revenue and cost components
- **Base vs Simulated Values Table** for analyzing the changes in the simulation values as compared to the base values across the PBT, recovery, revenue and cost components
- **Share Table** for analyzing the simulated shares vs the base shares
- **Summary Cards** for Key KPIs
- **Dynamic Text Card** for showing change in PBT% and the highest share segment

- **Page 11: PA Mix:**

- **Sliders** for adjusting mix values for **PA/Non-PA** type of loans
- **Simulated Values Table** for viewing the changes in simulated values across the loan types for PBT, recovery, revenue and cost components
- **Base vs Simulated Values Table** for analyzing the changes in the simulation values as compared to the base values across the PBT, recovery, revenue and cost components
- **Share Table** for analyzing the simulated shares vs the base shares
- **Summary Cards** for Key KPIs
- **Dynamic Text Card** for showing change in PBT% and the highest share loan type

- **Purpose of What-If Scenario:**

- Test business decisions before implementing them
- Understand which segments' combination yields the highest returns
- Align sourcing policies according to risk and return goals

The dashboard structure is designed for maximum usability and business relevance. Each page guides the user through a journey — from understanding the past, evaluating the present, and planning for the future — all in one integrated analytical environment.

The Power BI dashboard is formulated with a **modular and objective-wise structure** that aligns directly with the four business problems addressed during the internship. Each page of the

dashboard corresponds to one key business question and is equipped with dynamic filters and measures to allow for user-led exploration. The slicer panel has been provided on the right side of the dashboard, using which one can filter the record as per their needs and requirements. One can click on the specific button for easy navigation, title has been added on each button for understanding.

5. RESULTS AND VISUALS (*WITH DAX CODES*)

This section provides a detailed walkthrough of every Power BI dashboard module, explaining the purpose, components, key visuals, filters, and the DAX logic used to calculate profitability, risk, forecasting, and scenario simulations.

Each module aligns with a specific business objective defined during the project — ensuring that stakeholders can use the dashboard to take informed decisions.

By the end of the internship, a fully functional and highly dynamic **Power BI dashboard** was developed that enables one to:

- Monitor sourcing performance
- Evaluate customer profitability
- Identify risk patterns
- Simulate sourcing strategy changes
- Forecast future profitability

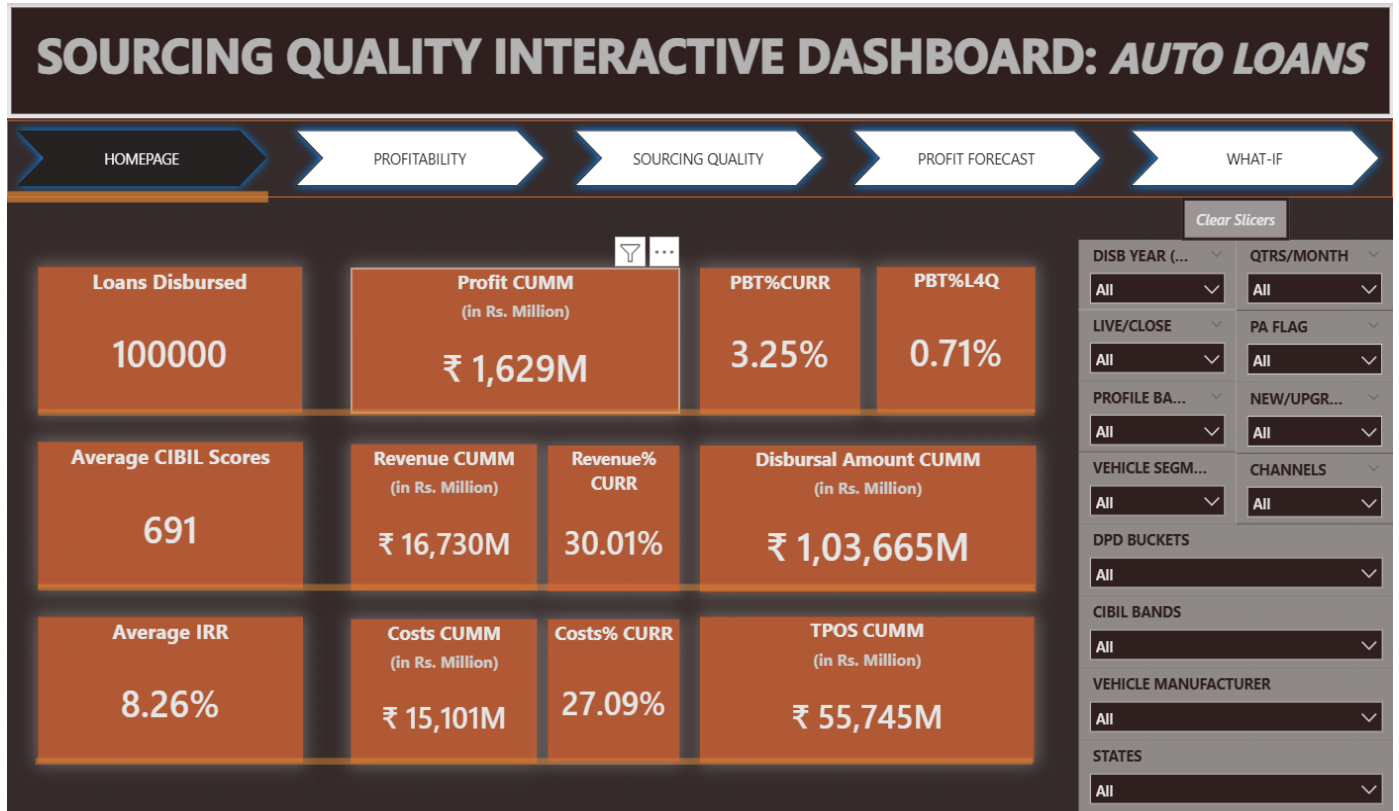
The dashboard provides actionable insights to business decision-makers and serves as a powerful analytical tool to support the bank's **retail lending strategy**.

Final Dashboard Snapshots (*With DAX codes*)

5.1 Homepage

- **KPIs with DAX**

- PBT% (Current, L4Q)
- Revenue% (Current)
- Costs% (Current)



- PBT% (Current)

●

●

●

PBT% (Current)

$$PBT\%Curr = ([ALMG\%Curr] + [BOC\%Curr] + [Fee\%Curr] + [Interest\%Curr] + [Recovery\%Curr]) - ([COF\%Curr] + [DRE\%Curr] + [DMA\%Curr] + [OPEX\%Curr] + [CC\%Curr])$$

(DAX for each will be covered in further pages)

- PBT% (L4Q)

●

●

●

PBT% (L4Q)

$$PBT\%L4Q = ([Recovery\%L4Q] + [Fee\%L4Q] + [BOC\%L4Q] + [Interest\%L4Q] + [ALMG\%L4Q]) - ([DMA\%L4Q] + [DRE\%L4Q] + [COF\%L4Q] + [CC\%L4Q] + [OPEX\%L4Q])$$

- Revenue% (Current)



Revenue% (Current)

$$\text{Revenue}\% \text{Curr} = [\text{Interest}\% \text{Curr}] + [\text{ALMG}\% \text{Curr}] + [\text{BOC}\% \text{Curr}] + [\text{Fee}\% \text{Curr}]$$

- **Costs% (Current)**



Costs% (Current)

$$\text{Costs}\% \text{Curr} = [\text{COF}\% \text{Curr}] + [\text{OPEX}\% \text{Curr}] + [\text{DRE}\% \text{Curr}] + [\text{DMA}\% \text{Curr}] + [\text{CC}\% \text{Curr}]$$

5.2 Profitability Analysis

- **DAX derived KPIs:**

- Profit per loan (L4Q)
- Revenue% (L4Q)
- Costs% (L4Q)
- COF% (L4Q)
- OPEX% (L4Q)
- DRE% (L4Q)
- DMA% (L4Q)
- CC% (L4Q)
- ALMG% (L4Q)
- BOC% (L4Q)
- Fee% (L4Q)
- Interest% (L4Q)
- Recovery% (L4Q)



- Profit per loan (L4Q)

Profit per loan (L4Q)

```
Profit_per_Loan(L4Q) = SUM(Loan_fact_table[Profits_L4Q])/COUNT(Loan_fact_table[LOAN_NO])
```

- Revenue% (L4Q)

Revenue% (L4Q)

```
Revenue%L4Q = [Interest%L4Q]+[Fee%L4Q]+[BOC%L4Q]+[ALMG%L4Q]
```

- Costs% (L4Q)

Costs% (L4Q)

```
Costs%L4Q = [COF%L4Q]+[OPEX%L4Q]+[DRE%L4Q]+[DMA%L4Q]+[CC%L4Q]
```

- COF% (L4Q)

COF% (L4Q)

$$\text{COF\%L4Q} = \frac{\text{SUM}(\text{Loan_fact_table}[\text{TOT_COST_COF_L4Q}])}{\text{SUM}(\text{Loan_fact_table}[\text{TP05}])}$$

The same logic works for DRE%, DMA%, OPEX%, CC%, Fee%, Interest%, ALMG%, BOC% and Recovery% L4Q.

5.3 Sourcing Quality and Default Analysis:

- **DAX derived KPIs:**
 - Bad Rate 30+ @12MOB
 - Bad Rate 90+ @12MOB
 - Bad Rate 30+ @24MOB
 - Bad Rate 90+ @24MOB
 - Default Rate
 - NPA Rate

SOURCING QUALITY AND DEFAULT ANALYSIS

Home

NEXT

BACK

TOTAL LOANS
DISBURSED

100000

AVERAGE CIBIL

691

AVERAGE IRR

8.26%

AVERAGE DPD

5

BAD RATES All

TRENDS

● Default Rate ● NPL Rate ● BadRate30+@12MOB ● BadRate30+@24MOB ● BadRate90+@12MOB ● BadRate90+@24MOB

BAD RATES CHANNEL_NEW

CHANNEL_NEW	Default_Rate_CURRENT	NPA_Rate_CURRENT	BadRate30+@12MOB	BadRate30+@24MOB	BadRate90+@12MOB	BadRate90+@24MOB
BRANCH	1.40%	0.54%	0.52%	0.70%	0.20%	0.38%
OAP	1.61%	0.56%	0.57%	1.03%	0.22%	0.30%
DSA	2.34%	0.77%	0.93%	1.17%	0.33%	0.60%
Total	1.99%	0.69%	0.78%	1.00%	0.28%	0.52%

DISB YEAR (...)

All

QTRS/MONTH

All

LIVE/CLOSE

All

PA FLAG

All

PROFILE BA...

All

NEW/UPGR...

All

VEHICLE SEG...

All

CHANNELS

All

DPD BUCKETS

All

CIBIL BANDS

All

VEHICLE MANUFACTURER

All

STATES

All

Clear Slicers

- **Bad Rate 30+ @12MOB**

```

Bad Rate 30+@12MOB

BadRate30+@12MOB =
DIVIDE(
    SUM( 'Loan_fact_table' [30+_12MOB_Settlement] ), SUM( 'Loan_fact_table' [DISBURSAL_AMT] )
)

```

Same logic follows for all other Bad Rates

- **Default Rate**

```

Default Rate%

Default_Rate =
DIVIDE( CALCULATE( SUM( 'Loan_fact_table' [TPOS] ), 'Loan_fact_table' [DPD] > 0 ),
    SUM( 'Loan_fact_table' [DISBURSAL_AMT] )
)

```

- **NPA Rate**

```

NPA Rate%

NPA_Rate =
DIVIDE( CALCULATE( SUM( 'Loan_fact_table' [TPOS] ), 'Loan_fact_table' [DPD] > 90 ),
    SUM( 'Loan_fact_table' [DISBURSAL_AMT] )
)

```

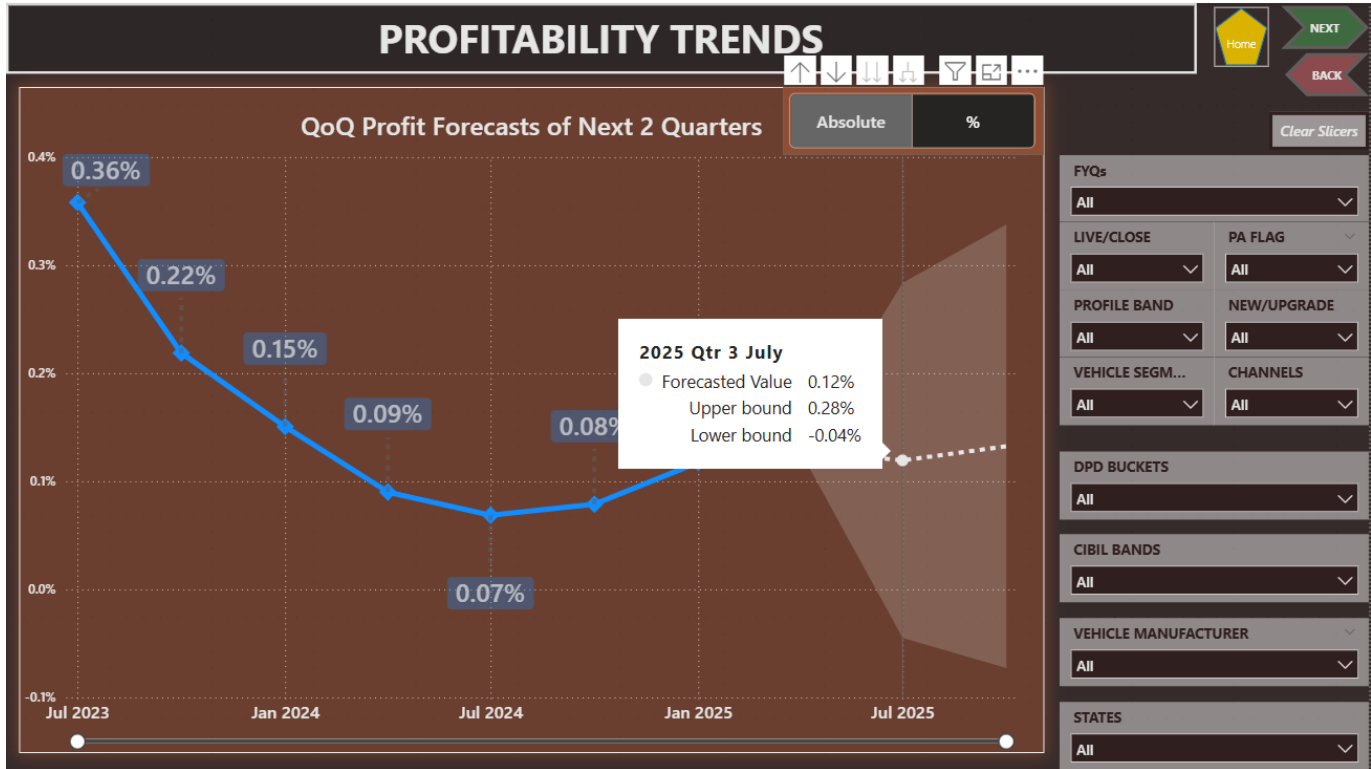
5.4 Future Profitability Forecasting

- Uses previous PBT data (QoQ) to forecast future profitability trends
- Time-series analysis allows for quarterly visualization of profit
- Aids strategic planning based on current performance trends

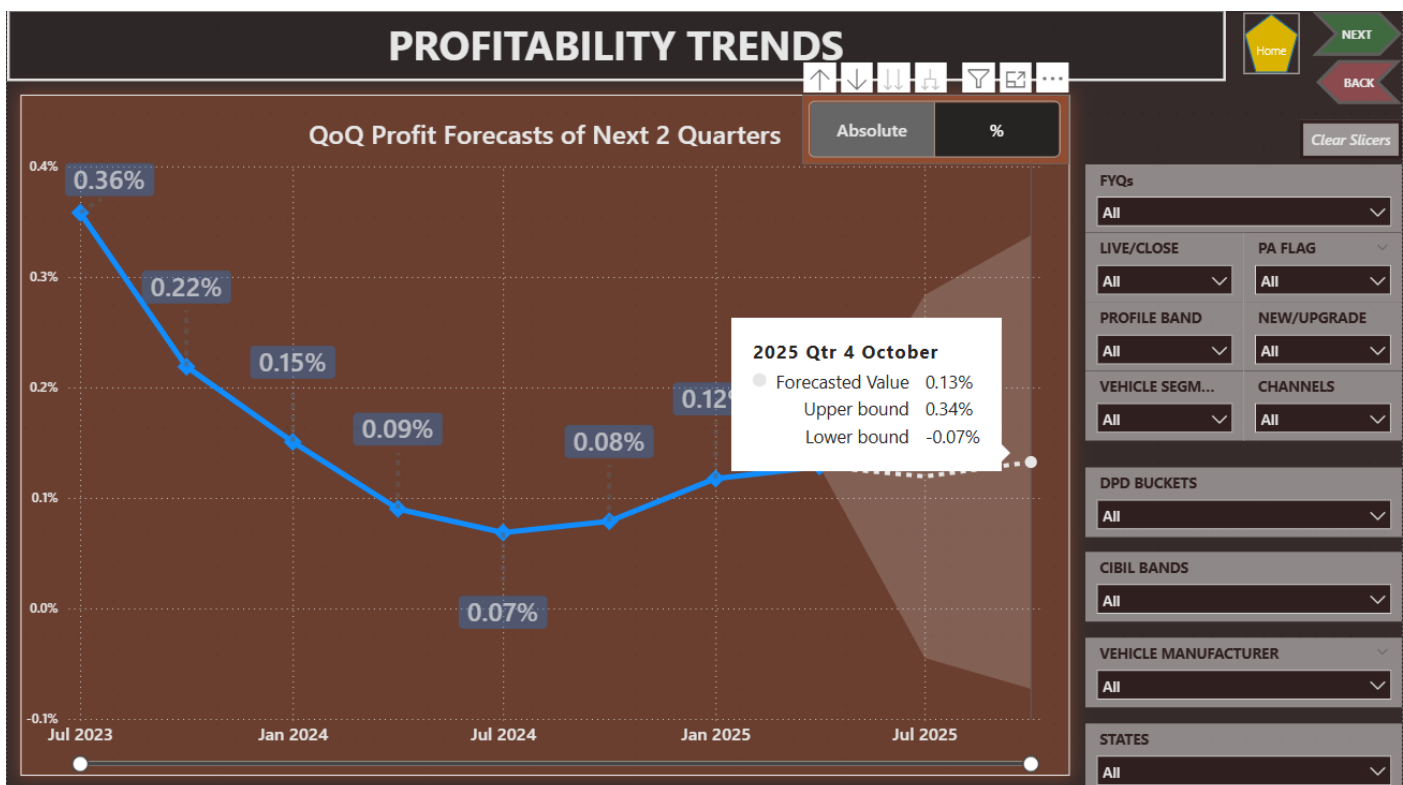
Note: The forecasting is done via the in-built model of Power BI i.e., ETS (Exponential Triple Smoothing) which allows Level, Trend and Seasonality as a factor for predicting future values.

- **DAX for Forecasting:**

- Forecasting DATES
- Quarterly Profit (Absolute)
- Quarterly Profit (%)



(Profit % Forecast: FY26 Q1)



(Profit % Forecast: FY26 Q2)

- **Forecasting Dates:**

```

Forecasting Dates

forecasting_dates =
ADDCOLUMNS(
    SELECTCOLUMNS(
        {
            ("FY24Q1", 2023, 7, 1),
            ("FY24Q2", 2023, 10, 1),
            ("FY24Q3", 2024, 1, 1),
            ("FY24Q4", 2024, 4, 1),
            ("FY25Q1", 2024, 7, 1),
            ("FY25Q2", 2024, 10, 1),
            ("FY25Q3", 2025, 1, 1),
            ("FY25Q4", 2025, 4, 1)
        },
        "FYQ", [Value1],
        "Year", [Value2],
        "Month", [Value3],
        "Day", [Value4]
    ),
    "StartDate", DATE([Year], [Month], [Day])
)

```

- **Quarterly Profit Absolute & (%)**

```

Quarterly Profit (Absolute)

Quarterly_Profit =
SWITCH(
    SELECTEDVALUE('forecasting_dates'[FYQ]),
    "FY24Q1", [Profit_FY24Q1],
    "FY24Q2", [Profit_FY24Q2],
    "FY24Q3", [Profit_FY24Q3],
    "FY24Q4", [Profit_FY24Q4],
    "FY25Q1", [Profit_FY25Q1],
    "FY25Q2", [Profit_FY25Q2],
    "FY25Q3", [Profit_FY25Q3],
    "FY25Q4", [Profit_FY25Q4]
)

```

```

Quarterly Profit (%)

Quarterly_Profit% =
SWITCH(
    SELECTEDVALUE('forecasting_dates'[FYQ]),
    "FY24Q1", [Profit_FY24Q1%],
    "FY24Q2", [Profit_FY24Q2%],
    "FY24Q3", [Profit_FY24Q3%],
    "FY24Q4", [Profit_FY24Q4%],
    "FY25Q1", [Profit_FY25Q1%],
    "FY25Q2", [Profit_FY25Q2%],
    "FY25Q3", [Profit_FY25Q3%],
    "FY25Q4", [Profit_FY25Q4%]
)

```

5.5 What-If Scenario (Unified Simulation)

- **DAX for simulations:**

- Base Values

- Simulated Values
- CIBIL Base and Simulated Shares
- New/Upgrade Base and Simulated Shares
- Profile Bands Base and Simulated Shares
- Segments Base and Simulated Shares
- PA Flag Base and Simulated Shares
- Channel Base and Simulated Shares
- Average IRR% Simulated (Unified)
- CC% Simulated (Unified)
- DMA% Simulated (Unified)
- DRE% Simulated (Unified)
- OPEX% Simulated (Unified)
- Interest% Simulated (Unified)
- Fee% Simulated (Unified)
- Recovery% Simulated (Unified)
- PBT% Simulated (Unified)
- Dynamic Input Validation Card

UNIFIED SIMULATION

Home

BACK

CIBIL BANDS (Click Here)

-1 (No History)

8.70

<700

5.40

700-724

7.50

725-749

18.90

750-779

35.00

780-799

19.90

>800

4.60

PROFILE BANDS (Click Here)

SALARIED

23.70

NEW/UPGRADE (Click Here)

NEW

66.70

Segments (Click Here)

Luxury

1.20

CHANNELS (Click Here)

BRANCH

31.30

DSA

66.80

OAP

1.90

PA FLAG (Click Here)

PA

19.00

Valid Input

RESET

Individual What-IF

BASE VS SIMULATED VALUES

Metrics	Base_Values	Sim_Values	Change
Avg_IRR_L4Q %	8.26%	8.26%	-0.00%
CC_L4Q %	0.76%	0.76%	-0.00%
COF_L4Q %	7.07%	7.07%	0.00%
DMA_L4Q %	0.49%	0.49%	0.00%
DRE_L4Q %	0.04%	0.04%	-0.00%
Fee_L4Q %	0.29%	0.29%	0.00%
Interest_L4Q %	8.47%	8.47%	0.00%
OPEX_L4Q %	0.95%	0.95%	0.00%
PBT_L4Q %	0.71%	0.71%	0.00%
Recovery_L4Q %	0.18%	0.18%	-0.00%

CIBIL BANDS

CIBIL_Bands	Base_Mix	Sim_Mix
<700	5.43%	5.42%
>800	4.58%	4.57%
-1 (No History)	8.69%	8.68%
700-724	7.51%	7.50%
725-749	18.94%	18.92%
750-779	34.98%	34.93%
780-799	19.87%	19.97%
Total	100.00%	100.00%

NEW/UPGRADE

NEW/UPG	Base_Mix	Sim_Mix
NEW	66.70%	66.70%
UPGRADE	33.30%	33.30%
Total	100.00%	100.00%

PROFILE BANDS

PROFILES	Base_Mix	Sim_Mix
SAL	23.72%	23.70%
SE+OTHERS	76.29%	76.30%
Total	100.00%	100.00%

SEGMENTS

Segments	Base_Mix	Sim_Mix
Luxury	1.20%	1.20%
Non-Luxury	98.80%	98.80%
Total	100.00%	100.00%

PA FLAG

PA/NPA	Base_Mix	Sim_Mix
NON PA	81.01%	81.00%
PA	18.99%	19.00%
Total	100.00%	100.00%

CHANNEL

CHANNELS	Base_Mix	Sim_Mix
BRANCH	31.32%	31.32%
DSA	66.77%	66.77%
OAP	1.91%	1.91%
Total	100.00%	100.00%

- Base Values and Simulated Values

```

Base Values

Base_Values =
SWITCH(
    SELECTEDVALUE(MetricList[Metrics]),
    "Interest_L4Q %", [Interest%L4Q],
    "Fee_L4Q %", [Fee%L4Q],
    "BOC_L4Q %", [BOC%L4Q],
    "ALMG_L4Q %", [ALMG%L4Q],
    "Recovery_L4Q %", [Recovery%L4Q],
    "COF_L4Q %", [COF%L4Q],
    "CC_L4Q %", [CC%L4Q],
    "DMA_L4Q %", [DMA%L4Q],
    "DRE_L4Q %", [DRE%L4Q],
    "OPEX_L4Q %", [OPEX%L4Q],
    "PBT_L4Q %", [PBT%L4Q],
    "Avg_IRR_L4Q%", AVERAGE('Loan_fact_table'[IRR%])
)

```

```

Simulated Values

Simulated_Values =
SWITCH(
    SELECTEDVALUE(MetricList[Metrics]),
    "Interest_L4Q %", [Interest_Uni],
    "Fee_L4Q %", [Fee_uni],
    "BOC_L4Q %", [BOC%L4Q],
    "ALMG_L4Q %", [ALMG%L4Q],
    "Recovery_L4Q %", [Recovery_Uni],
    "COF_L4Q %", [COF%L4Q],
    "CC_L4Q %", [CC_uni],
    "DMA_L4Q %", [DMA_Uni],
    "DRE_L4Q %", [DRE_Uni],
    "OPEX_L4Q %", [OPEX_uni],
    "PBT_L4Q %", [PBT_Uni],
    "Avg_IRR_L4Q%", [IRR_uni]
)

```

- Average IRR (%) Simulated

```

IRR% Simulation (Unified)

IRR_uni = ([Avg_IRR%_Band]+[Avg_IRR%_Cust]+[Avg_IRR%_PA]+[Avg_IRR%_Segment]+[Avg_IRR%_Sim]+
[Sim_IRR%CIBIL])/[TotalLoans]*100

```

- CC% Simulated

```

Credit Cost% Simulation (Unified)

CC_uni = ([CC%Channel]+[CC%CIBIL]+[CC%NEW/UPG]+[CC%PA]+[CC%Profile]+[CC%Segment])/[TotalLoans]*100

```

Rest of the components follow the same logic

- Dynamic Input Validation Card

```

Dynamic Input Validation Card

Input_Validation_Status =
VAR CIBIL_Total = SUMX(VALUES('Dim_Customer'[CIBIL_Bands]),[Sim_Share_CIBIL])

VAR Channel_Total = SUMX(VALUES('Dim_Channel'[CHANNEL_NEW]),[Sim_Share_Channel])

RETURN
IF(
    ABS(CIBIL_Total - 1) < 0.001 && ABS(Channel_Total - 1) < 0.001,
    "✅ Valid Input", "❌ Invalid Input: Total exceeds 100%"
)

```

- Channels Base Shares

Channel Share(%) Base

```
Base_Channel_Mix =
DIVIDE(
    COUNTROWS('Loan_fact_table'),
    CALCULATE(COUNTROWS('Loan_fact_table'), REMOVEFILTERS('Dim_Channel'[CHANNEL_NEW]))
)
```

Same logic follows for CIBIL Band, PA, New/Upgrade, Segments and Profiles' Base Shares

- **Simulated Shares Channel:**

Channel Share(%) Simulation

```
Sim_Share_Channel =
VAR Band = SELECTEDVALUE('Dim_Channel'[CHANNEL_NEW])
VAR Base_DSA = 66.77
VAR Base_BRANCH = 31.32
VAR Base_OAP = 1.91
VAR Input_DSA = [DSA_Mix Value]
VAR Input_BRANCH = [BRANCH_Mix Value]
VAR Input_OAP = [OAP_Mix Value]
VAR ChannelTable = ADDCOLUMNS(
    SELECTCOLUMNS(
        {
            ("DSA", Input_DSA, Base_DSA),
            ("BRANCH", Input_BRANCH, Base_BRANCH),
            ("OAP", Input_OAP, Base_OAP)
        },
        "Channel", [Value1],
        "Input", [Value2],
        "Base", [Value3]
    ),
    "IsChanged", IF(ABS([Input] - [Base]) > 0.1, TRUE, FALSE)
)
VAR ThisChannelRow = FILTER(ChannelTable, [Channel] = Band)
VAR ThisBase = MAXX(ThisChannelRow, [Base])
VAR ThisInput = MAXX(ThisChannelRow, [Input])
VAR IsThisChanged = IF(COUNTROWS(FILTER(ThisChannelRow, [IsChanged] = TRUE)) > 0, TRUE, FALSE)
VAR ChangedChannels = FILTER(ChannelTable, [IsChanged] = TRUE)
VAR UnchangedChannels = FILTER(ChannelTable, [IsChanged] = FALSE)
VAR SimulatedShare = IF(
    IsThisChanged,
    ThisInput,
    VAR TotalChangedInput = SUMX(ChangedChannels, [Input])
    VAR RemainingShare = 100 - TotalChangedInput
    VAR BaseUnchanged = SUMX(UnchangedChannels, [Base])
    RETURN IF(
        BaseUnchanged > 0,
        IF(
            RemainingShare > 0,
            DIVIDE(ThisBase, BaseUnchanged) * RemainingShare,
            DIVIDE(ThisBase, BaseUnchanged) * 5
        ),
        0
    )
)
VAR IsTotal = NOT(ISINSCOPE('Dim_Channel'[CHANNEL_NEW]))
RETURN IF(IsTotal, 1, ROUND(SimulatedShare / 100, 4))
```

Same logic follows for CIBIL Band, PA, New/Upgrade, Segments and Profiles' Simulated Shares

Logic applied for normalized share simulation: (followed in all simulations)

If Sourcing from **Branch** is increased, from **31.32% to 40%**

<u>Channels</u>	<u>Base Share</u>	<u>Increased to</u>
Branch	31.32%	40.00%
DSA	66.77%	
OAP	1.91%	
<u>Total</u>	100.00%	

Remaining % to distribute	60.00%	(100% - 40%)
Remaining Base	68.68%	(66.77% + 1.91%)

DSA Sim = DSA Base * (Remaining% to distribute / Remaining Base)	66.77% * (60% / 68.68%) =	58.33%
OAP Sim = OAP Base * (Remaining % to distribute / Remaining Base)	1.91% * (60% / 68.68%) =	1.67%

If Sourcing from **Branch** is decreased to **20%** and **DSA** is decreased to **30%**

<u>Channels</u>	<u>Base Share</u>	<u>Increased to</u>
Branch	31.32%	20.00%
DSA	66.77%	30.00%
OAP	1.91%	
<u>Total</u>	100.00%	

Remaining % to distribute	50.00%	(100% - (20%+30%))
Remaining Base	1.91%	

OAP Sim = OAP Base * (Remaining % to distribute / Remaining Base)	1.91% * (50% / 1.91%) =	50.00%
---	-------------------------	---------------

5.6 What-IF Scenario (Individual Simulation)

• 5.6.1 Channel Simulation

- Channels' Base Shares (*already covered*)
- Channels' Simulated Shares (*already covered*)
- Channels' IRR% Simulated
- Channels' Interest% Simulated
- Channels' Fee% Simulated
- Channels' CC% Simulated
- Channels' DMA% Simulated
- Channels' DRE% Simulated
- Channels' OPEX% Simulated
- Channels' Recovery% Simulated
- Channels' PBT% Simulated

```

Channels' IRR% Simulation

Avg_IRR%_Sim =
VAR Channel = SELECTEDVALUE( 'Dim_Channel' [CHANNEL_NEW] )

VAR Channelwise_IRR =
    SWITCH(
        Channel,
        "BRANCH", [Avg_IRR_BRANCH],
        "DSA", [Avg_IRR_DSA],
        "OAP", [Avg_IRR_OAP]
    )

VAR Total_IRR =
    [BRANCH_Mix Value] * [Avg_IRR_BRANCH] +
    [DSA_Mix Value] * [Avg_IRR_DSA] +
    [OAP_Mix Value] * [Avg_IRR_OAP]

VAR Total_Loans =
    [BRANCH_Mix Value] +
    [DSA_Mix Value] +
    [OAP_Mix Value]

RETURN
IF(
    HASONEVALUE( 'Dim_Channel' [CHANNEL_NEW] ),
    Channelwise_IRR,
    DIVIDE(Total_IRR, Total_Loans)
)

```

- **Channel's Interest% Simulation**

Channels' Interest Income% Simulation

```

INTEREST%Channel =
VAR Band = SELECTEDVALUE('Dim_Channel'[CHANNEL_NEW])
VAR Sim_INTEREST =
    SWITCH(
        Band,
        "BRANCH", DIVIDE([BRANCH_Mix Value] * [Avg_INTEREST_BRANCH], [BRANCH_Mix Value] *
[Avg_TPOS_BRANCH]),
        "DSA", DIVIDE([DSA_Mix Value] * [Avg_INTEREST_DSA], [DSA_Mix Value] * [Avg_TPOS_DSA]),
        "OAP", DIVIDE([OAP_Mix Value] * [Avg_INTEREST_OAP], [OAP_Mix Value] * [Avg_TPOS_OAP])
    )
VAR Total_INTEREST =
    [BRANCH_Mix Value] * [Avg_INTEREST_BRANCH] +
    [DSA_Mix Value] * [Avg_INTEREST_DSA] +
    [OAP_Mix Value] * [Avg_INTEREST_OAP]
VAR Total_TPOS =
    [BRANCH_Mix Value] * [Avg_TPOS_BRANCH] +
    [DSA_Mix Value] * [Avg_TPOS_DSA] +
    [OAP_Mix Value] * [Avg_TPOS_OAP]
RETURN
    IF(HASONEVALUE('Dim_Channel'[CHANNEL_NEW]), Sim_INTEREST, DIVIDE(Total_INTEREST, Total_TPOS))

```

The rest of the Individual Simulations and their components follow the same logic.

- 5.6.2 CIBIL Bands Simulation

-1 (No History) Mix (%)

8.70

<700 Mix (%)

5.40

700-724 Mix (%)

7.50

725-749 Mix (%)

18.90

750-779 Mix (%)

35.00

780-799 Mix (%)

19.90

>800 Mix (%)

4.60

RESET

PBT%L4Q

0.71%

PBT% SIMULATED

0.71%

REVENUE% L4Q

9.83%

REVENUE% SIMULATED

9.83%

COSTS% L4Q

9.30%

COSTS% SIMULATED

9.30%

WHAT-IF SIMULATION- (CIBIL MIX)

Home

CIBIL_BandS

Base_Mix%

Sim_Mix%

<700	5.43%	5.42%
>800	4.58%	4.57%
-1 (No History)	8.69%	8.68%
700-724	7.51%	7.50%
725-749	18.94%	18.92%
750-779	34.98%	34.93%
780-799	19.87%	19.97%
Total	100.00%	100.00%

PBT changed by 0.00% and the highest share is from the 750-779 CIBIL band.

Valid Input

NEXT

BACK

BACK (UNIFIED)

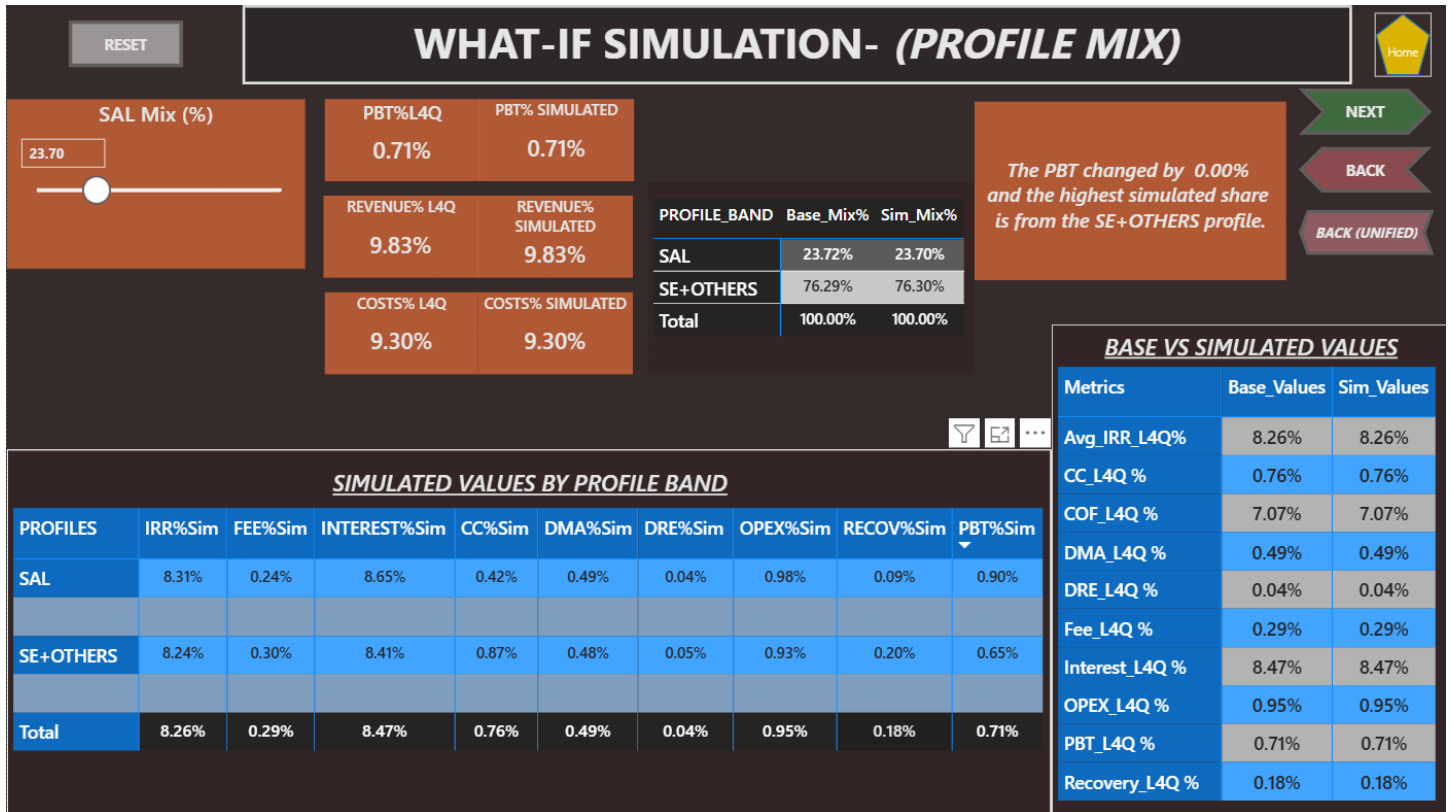
BASE VS SIMULATED VALUES

Metrics	Base_Values	Sim_Values
Avg_IRR_L4Q%	8.26%	8.26%
CC_L4Q %	0.76%	0.76%
COF_L4Q %	7.07%	7.07%
DMA_L4Q %	0.49%	0.49%
DRE_L4Q %	0.04%	0.04%
Fee_L4Q %	0.29%	0.29%
Interest_L4Q %	8.47%	8.47%
OPEX_L4Q %	0.95%	0.95%
PBT_L4Q %	0.71%	0.71%
Recovery_L4Q %	0.18%	0.18%

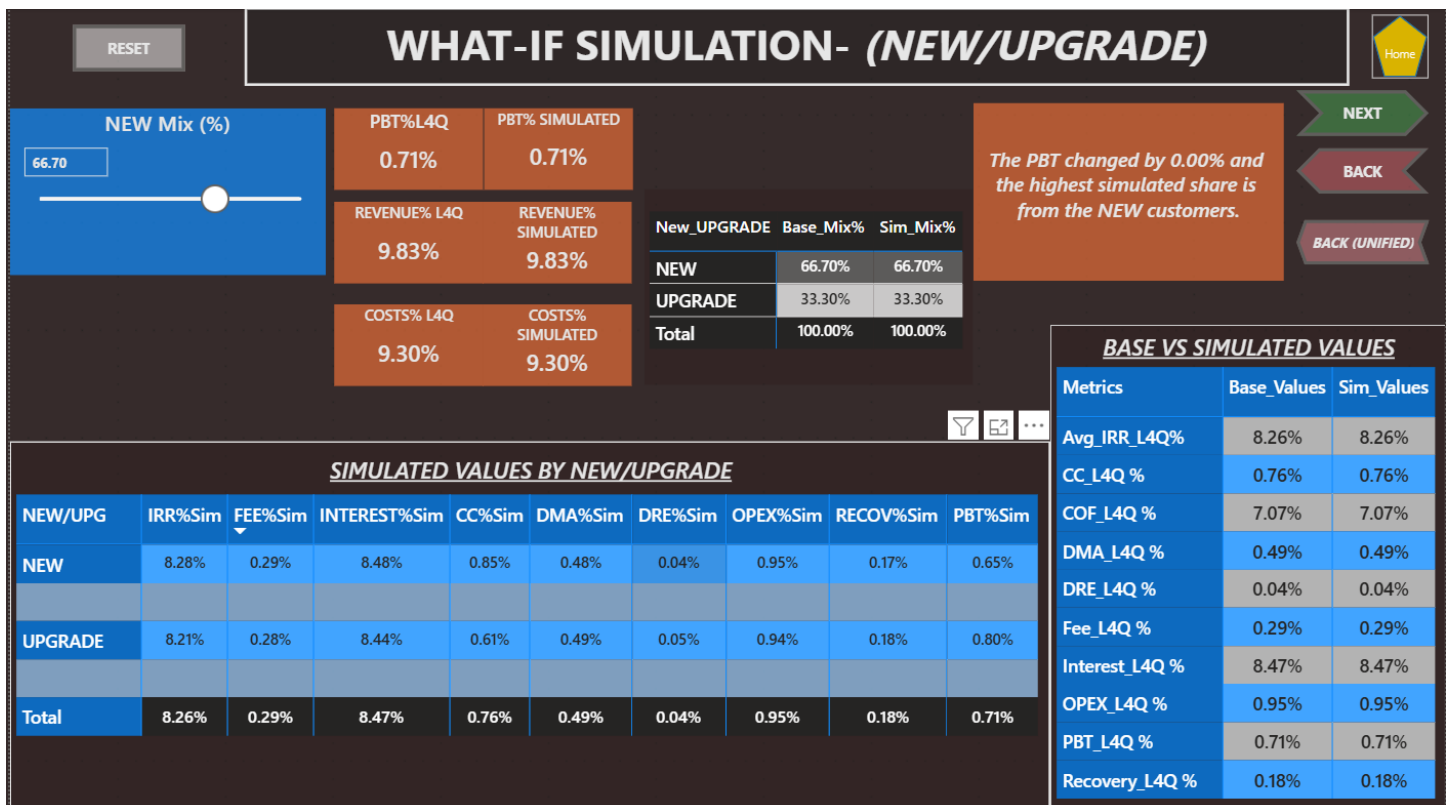
SIMULATED VALUES BY CIBIL BANDS

CIBIL_BandS	IRR% Sim	Fee% Sim	Interest %Sim	CC% Sim	DMA %Sim	DRE %Sim	OPEX %Sim	Recov% Sim	PBT% Sim
<700	8.49%	0.21%	9.51%	0.74%	0.52%	0.02%	1.06%	0.46%	1.34%
>800	7.89%	0.27%	8.15%	0.22%	0.49%	0.05%	0.92%	0.05%	0.89%
-1 (No History)	8.63%	0.30%	9.41%	1.05%	0.54%	0.03%	1.04%	0.27%	0.93%
700-724	8.52%	0.28%	8.98%	1.54%	0.51%	0.03%	0.99%	0.42%	0.35%
725-749	8.33%	0.32%	7.97%	1.05%	0.46%	0.06%	0.88%	0.19%	0.41%
750-779	8.26%	0.28%	8.57%	0.70%	0.49%	0.04%	0.96%	0.16%	0.75%
780-799	7.95%	0.29%	8.29%	0.35%	0.48%	0.05%	0.94%	0.06%	0.85%
Total	8.26%	0.29%	8.47%	0.76%	0.49%	0.04%	0.95%	0.18%	0.71%

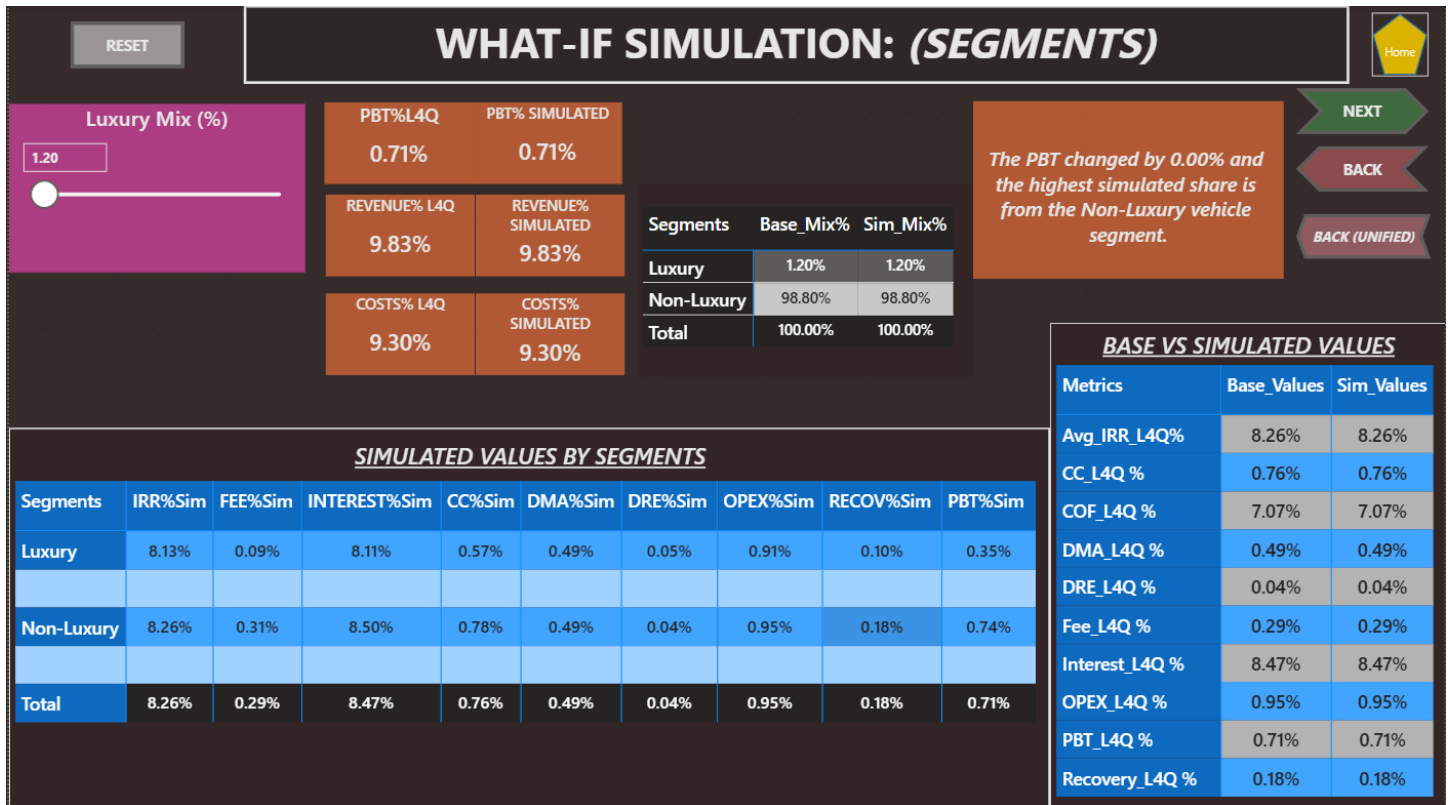
- 5.6.3 Profile Bands Simulation



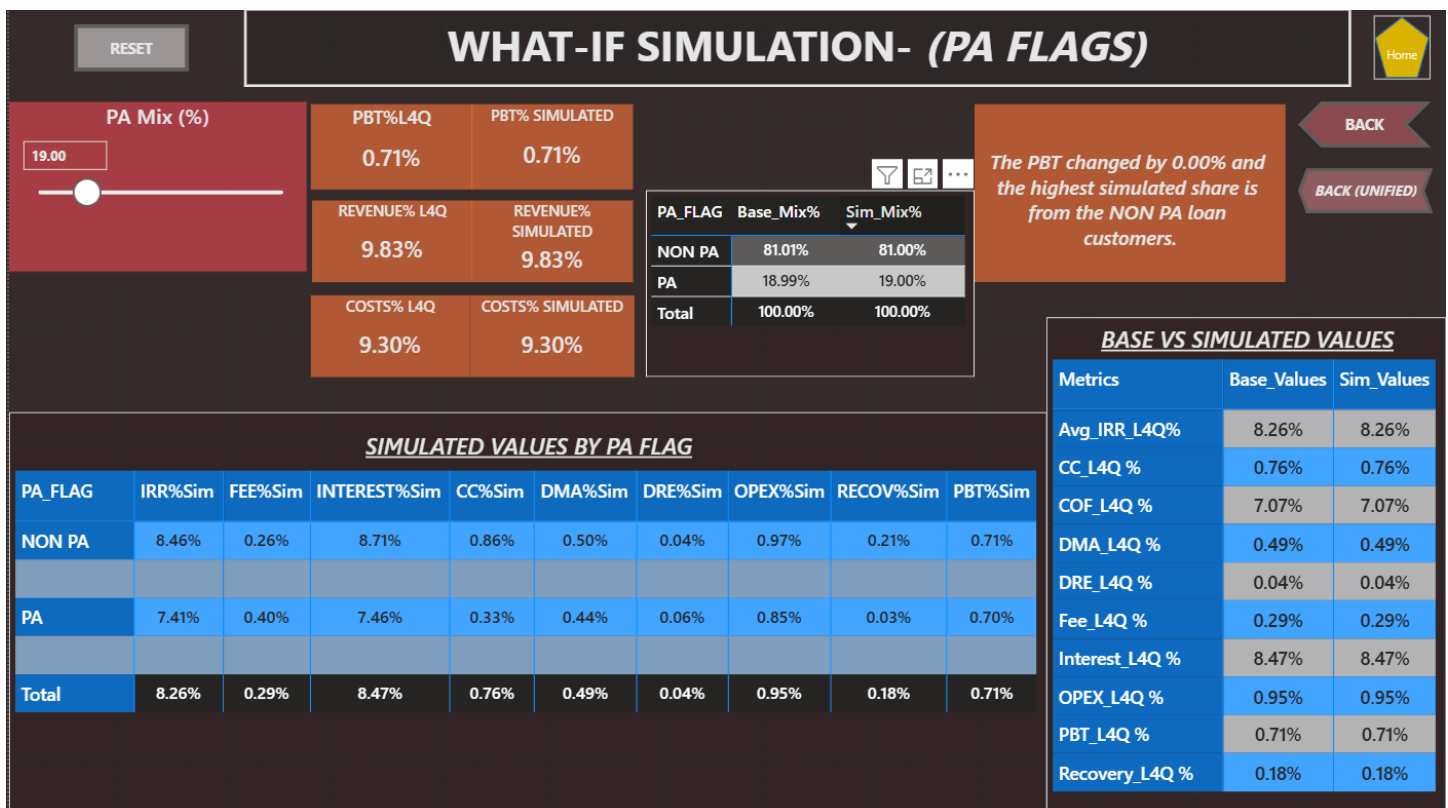
- 5.6.4 New/Upgrade Simulation



- 5.6.5 Vehicle Segments Simulation



- 5.6.6 PA/Non PA loans Simulation



6. LEARNING OUTCOMES AND CONCLUSION

The two-month internship at ICICI Bank Ltd. under the DSAG (Data Science and Analytics Group) proved to be a highly enriching and transformative experience. This project not only enhanced my technical competencies but also contributed to the development of professional and soft skills crucial for success in the analytics domain.

Key learning outcomes from the internship are as follows:

- **Advanced Data Visualization Skills:**

Gained hands-on experience in building professional dashboards using **Microsoft Power BI**. Learned how to create interactive visualizations, build complex **DAX measures**, and implement slicer-driven interactivity to support real-time business insights.

- **Data Cleaning and Structuring:**

Developed skills in working with large datasets, especially with respect to **data cleaning**, **logical consistency checks**, and **structuring simulated datasets** to resemble real-world banking data. **Microsoft Excel** was used extensively for preprocessing and validation.

- **KPI Modeling & Business Analytics:**

Understood how to define and calculate key performance indicators like **Profit Before Tax (PBT)**, **Bad Rate**, and other credit risk metrics, and how to derive insights based on sourcing channel, customer profile, and product type.

- **Forecasting and Simulation Techniques:**

Implemented **time series forecasting** to project future profitability trends, and developed a detailed **What-If Simulation model** to evaluate strategic changes in sourcing mix, thereby adding predictive power to the dashboard.

- **Soft Skills & Communication:**

Regular review meetings and discussions with the mentor and buddy helped improve my **communication, problem-solving, and critical thinking** skills. Learned how to accept feedback constructively and convert it into actionable improvements.

- **Foundation as a Data Analyst:**

This internship served as a strong foundation for building a career in **data analytics and business intelligence**, having given a clearer understanding of how real-world data is processed, interpreted, and used to guide critical business decisions.

Overall, the internship was a great blend of technical learning and professional development, equipping me with both the hard and soft skills required in the present data-driven industry.

The two-month summer internship at **ICICI Bank Ltd.** under the **Data Science and Analytics Group (DSAG)** provided a valuable and practical learning experience in the field of data analytics, business intelligence, and financial analysis.

Through the project titled “**Sourcing Quality Interactive Dashboard Preparation – Retail**”, I was able to apply analytical thinking, technical tools, and business logic to solve real-world banking problems using a data-driven approach. The project involved the design and development of an interactive **Power BI dashboard** that helped visualize key insights on **profitability** while working with simulated data and building dynamic dashboards equipped me with a deep understanding of:

- **Retail loan performance metrics**
- **Sourcing channel efficiency**
- **Customer risk segmentation**
- And the practical use of **Data Visualization tools like Power BI and Excel**

The internship also gave me exposure to:

- **Business communication** through review meetings and regular discussions with mentor and buddy
- **Technical challenges** like DAX optimization, slicer logic, and model consistency
- **Real-world thinking**, including how strategic sourcing decisions affect long-term profitability

*This internship has not only strengthened my technical foundation in **data analytics**, but also built a structured mindset toward solving business problems. It has prepared me to take on real-life projects with confidence and has laid the groundwork for my future career as a **Data Analyst or related profession**.*

PROJECT SUBMISSION: [OneDrive Link](#)
